1. **There are four relationships:**

Student(sid, sname, did, sex, birthdate, gpa)

Course(cid, cname, preid, did, semester)

/*pried indicates the course number of the pre-taken course*/

Enroll(sid, cid, grade)

Dept(did, dname, phdstunum)

/*phdstunum indicates the number of doctoral students in the department/

Write an SQL statement that expresses the following query requirements (must be expressed in SQL statements):

(1) Use the join query to find the names of students who scored 90 or above in more than one course in the fall semester.

(2) Use the nested query to find the names of students who scored 90 or above in more than one course in the fall semester.

(3) Use the nested query with correlation to find the names of students who scored 90 or above in more than one course in the fall semester.

(4) Select the name and the corresponding course number of the student whose grade of each course is not lower than the average score of the course.

(5) Query the cid, cname and average GPA of students for each course offered by the Computer Department.

2. **What are the differences between B Tree and B+ tree? Why do we use B+ tree in DBMS?**

3. **Explain why "OR" should be avoided in SQL WHERE clauses from the perspective of query optimization.**

4. **Compared with hierarchical and network database systems, query optimization is more important for relational database systems. Is that right? Why?**

5. **Read the instructions below and answer questions (1) and (2). Assume that the transactions T1 and T2 corresponding to the two businesses are related to deposit relationship:**

   Transfer businesses -- T1(A,B, S) , transfers S dollars from account A to account B;

   Interest calculation business -- T2, calculating interest for all current accounts (i.e., the original amount is X dollars, which is X*1.2 after interest calculation)

(1) **If the interest calculation business is designed to calculate interest on a single account separately, that is, T2(A) calculates interest on account A, and T2(B) calculates interest on account B. Is this scheme correct? Why ?**

(2) **Assume a concurrent scheduling of two transactions T1 and T2 as shown in the following table, <u>introduce (S, U, X) lock (note: the compatibility matrix is as shown in the figure below)</u>, if the initial A=100, B= 60, S=20, what is the final result of A and B? Is the concurrent scheduling correct?**

T1(A,B,s)	T2
Read(A)	
A:=A-s	Read(A)
Write(A)	A:=A*1.2
	Write(A)
	Read(B)
Read(B)	B:=B*1.2
B:=B+s	Write(B)
Write(B)	

<table>
<tr><td></td><td colspan="4">Locks already owned by other transactions</td></tr>
<tr><td rowspan="4">Lock request</td><td></td><td>S</td><td>U</td><td>X</td></tr>
<tr><td>S</td><td>Y</td><td>Y</td><td>N</td></tr>
<tr><td>U</td><td>Y</td><td>Y</td><td>Y</td></tr>
<tr><td>X</td><td>N</td><td>N</td><td>N</td></tr>
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